

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT _*CO-SC(MODEL-A) **Date: 03-01-26**
Time: 3 Hrs **Max. Marks: 300**

03-01-26_SR.IIT_STAR CO-SC(MODEL-A&APEX)_JEE MAIN_GTM-7_SYLLABUS

PHYSICS: Electrostatics, Capacitors, Current electricity, electro magnetism, EMI & AC, EM waves, Modern physics, Semiconductors & Electronic devices, Ray optics and Optical instruments, Wave optics

CHEMISTRY: Liquid solutions, Chemical kinetics, Electro chemistry, Haloalkanes & haloarenes, Alcohols, phenols & ethers, Aldehydes, ketones & carboxylic acids, Amines, Biomolecules, p-block, d and f-block, Coordination compounds, Principles related to practical chemistry

MATHEMATICS: Complex numbers, Quadratic expressions, Sequence & Series, Permutations and Combinations, Binomial theorem, Probability, Circle, Parabola, Ellipse, Hyperbola, Integration, Differential equations.

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PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. Four charges Q each are fixed at the vertices of a square ABCD. A particle of mass m and charge ' $-Q$ ' is released at rest at mid point of side AB. Select the correct options with respect its motion.

A) It stays at rest

B) It moves towards centre and attains a maximum speed at the centre

C) It moves towards centre but attains a local minima of speed at the centre

D) It will begin to move towards centre but will never reach the center

2. Match List-I (Electromagnetic wave type) with List - II (Its association/application) and select the correct option from the choices given below the lists:

List-I		List-II	
(a)	Infrared waves	(i)	To treat muscular strain
(b)	Radio waves	(ii)	For broadcasting
(c)	X-rays	(iii)	To detect fracture of bones
(d)	Ultraviolet rays	(iv)	Absorbed by the ozone layer of the atmosphere

A) a – iv, b – iii, c – ii, d – i

B) a – i, b – ii, c – iv, d – iii

C) a – iii, b – ii, c – i, d – iv

D) a – i, b – ii, c – iii, d – iv

3.

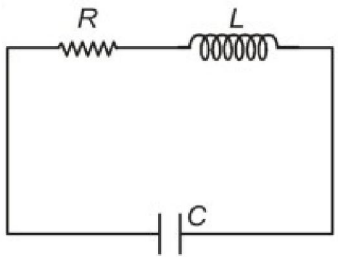


A point object o is observed through a thin film of thickness t and refractive index μ. observer is far from film and film is Normal to line joining o and observer. Choose the correct option regarding change in optical path without rays due to film compare to without film

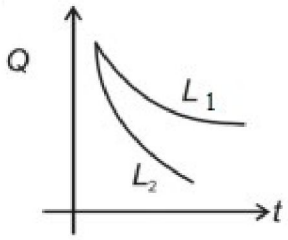
A) $t\left(1-\frac{1}{\mu}\right)$ B) $\frac{\mu t}{(\mu-1)}$ C) $\frac{t}{\mu-1}$ D) $t(\mu-1)$

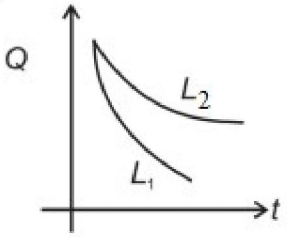
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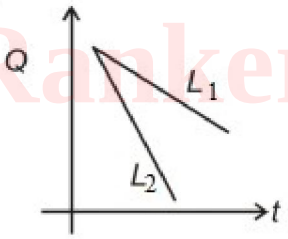
4. An LCR circuit is equivalent to a damped pendulum. In an LCR circuit the capacitor is charged to Q_0 and then connected to the L and R as shown below:

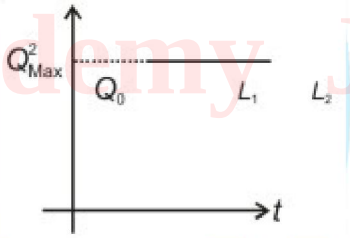


If a student plots graphs of the square of maximum charge (Q^2_{Max}) on the capacitor with time (t) for two different values L_1 and $L_2(L_1 > L_2)$ of L then which of the following represents this graph correctly? (Plots are schematic and not drawn to scale)

A) 

B) 

C) 

D) 

5. You are given four sources of light each one providing a light of a single colour - red, blue, green and yellow. Suppose the angle of refraction for a beam of yellow light corresponding to a particular angle of incidence at the interface of two media is 90°. Which of the following statements is correct if the source of yellow light is replaced with that of other lights without changing the angle of incidence?

A) The beam of red light would undergo total internal reflection.

B) The beam of red light would bend towards normal while it gets refracted through the second medium.

C) The beam of blue light would undergo total internal reflection.

D) The beam of green light would bend away from the normal as it gets refracted through the second medium.

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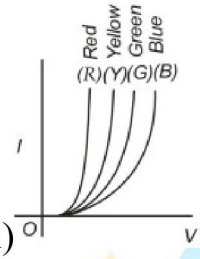
6. A parallel plate Capacitor of Capacitance C is connected across a battery of voltage V. If the plates are suddenly pulled apart to double the separation, work done by battery till steady state will be

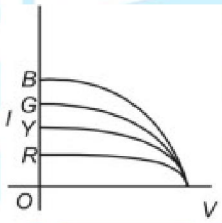
A) CV^2 B) $\frac{1}{2}CV^2$ C) 0 D) $-\frac{1}{2}CV^2$

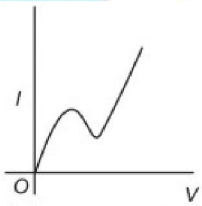
7. There are two long co-axial solenoids of same length l. The inner and outer coils have radii r_1 and r_2 and number of turns per unit length n_1 and n_2, respectively. The ratio of mutual inductance to the self inductance of the inner-coil is

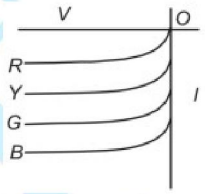
A) $\frac{n_2}{n_1}\cdot\frac{r_2^2}{r_1^2}$ B) $\frac{n_2}{n_1}$ C) $\frac{n_2}{n_1}\cdot\frac{r_1}{r_2}$ D) $\frac{n_1}{n_2}$

8. The $I-V$ characteristic of an LED is

A) 

B) 

C) 

D) 

9. For the ground state, the electron in the H -atom has an angular momentum $=\hbar$, according to the simple Bohr model. Angular momentum is a vector and hence there will be infinitely many orbits with the vector pointing in all possible directions. In actuality, this is not true,

A) because Bohr model gives incorrect values of angular momentum.
B) because only one of these would have a minimum energy.
C) angular momentum must be in the direction of spin of electron.
D) because electrons go around only in horizontal orbits.

10. For point charges $-q,+q,+q$ and $-q$ are placed on y-axis at $y=-2d, y=-d, y=+d$ and $y=+2d$, respectively. The magnitude of the electric field E at a point on the x-axis at $x=D$, with $D\gg d$, will behave as

A) $E\propto\frac{1}{D^3}$ B) $E\propto\frac{1}{D}$ C) $E\propto\frac{1}{D^4}$ D) $E\propto\frac{1}{D^2}$

11. **Statement-1**: The temperature dependence of resistance is usually given as $R=R_0(1+\alpha\Delta t)$. The resistance of a wire changes from 100Ω to 150Ω when its temperature is increased from $27^\circ C$ to $227^\circ C$. This implies that $\alpha=2.5\times10^{-3}/^\circ C$.

Statement-2: $R=R_0(1+\alpha\Delta t)$ is valid only when the change in the temperature ΔT is small and $\Delta R=(R-R_0)\ll R_0$.

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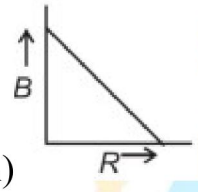
A) Statement 1 is true, statement 2 is true; Statement 2 is the correct explanation of Statement 1

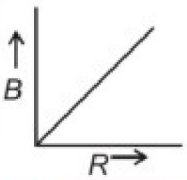
B) Statement 1 is true, Statement 2 is true; Statement 2 is not the correct explanation of Statement 1

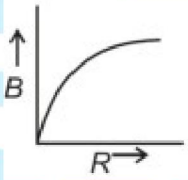
C) Statement 1 is false, Statement 2 is true

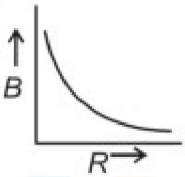
D) Statement 1 is true, Statement 2 is false

12. A charge Q is uniformly distributed over the surface of non-conducting disc of radius R. The disc rotates about an axis perpendicular to its plane and passing through its centre with an angular velocity ω. As a result of this rotation a magnetic field of induction B is obtained at the centre of the disc. If we keep both the amount of charge placed on the disc and its angular velocity to be constant and vary the radius of the disc then the variation of the magnetic induction at the centre of the disc will be represented by the figure

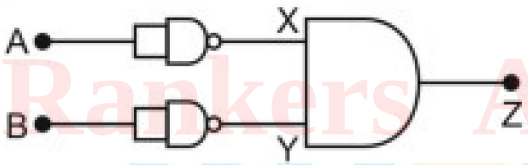
A) 

B) 

C) 

D) 

13. Identify the logic operation carried out by the given circuit:



A) NOR B) AND C) OR D) NAND

14. A hoop and a solid cylinder of same mass and radius are made of a permanent magnetic material with their magnetic moment parallel to their respective axes. But the magnetic moment of hoop is twice of solid cylinder. They are placed in a uniform magnetic field in such a manner that their magnetic moments make a small angle with the field. If the oscillation periods of hoop and cylinder are T_h and T_c respectively, then

A) $T_h = 0.5T_c$ B) $T_h = T_c$ C) $T_h = 2T_c$ D) $T_h = 1.5T_c$

15. Consider the two idealized systems: (i) a parallel plate capacitor with large plates and small separation and (ii) a long solenoid of length $L \gg R$, radius of cross-section. In (i) E is ideally treated as a constant between plates and zero outside. In (ii) magnetic field is constant inside the solenoid and zero outside. These idealised assumptions, however, contradict fundamental laws as below:

A) case (i) contradicts Gauss's law for electrostatic fields.

B) Case (ii) contradicts Gauss's law for magnetic fields.

C) case (i) agrees with $\oint \mathbf{E}.d\mathbf{l} = 0$

D) case (ii) contradicts $\oint \mathbf{H}.d\mathbf{l} = I_{en}$

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16. Consider a current carrying wire (current I) in the shape of a circle. Note that as the current progresses along the wire, the direction of j (current density) changes in an exact manner, while the current I remain unaffected. The agent that is essentially responsible for is

A) source of emf.

B) electric field produced by charges accumulated on the surface of wire.

C) the charges just behind a given segment of wire which push them just the right way by repulsion.

D) the charges ahead.

17. A coil is suspended in a uniform magnetic field, with the plane of the coil parallel to the magnetic lines of force. When a current is passed through the coil it starts oscillating; it is very difficult to stop. But if an aluminium plate is placed near to the coil, it stops. This is due to

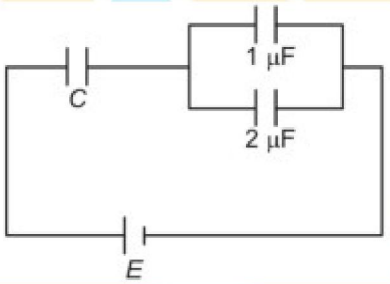
A) Induction of electrical charge on the plate

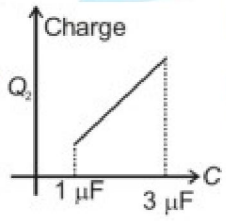
B) Shielding of magnetic lines of force as aluminium is a paramagnetic material

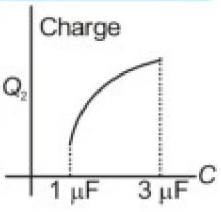
C) Electromagnetic induction in the aluminium plate giving rise to electromagnetic damping

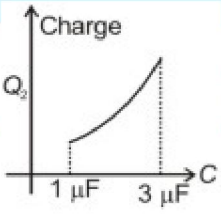
D) Development of air current when the plate is placed

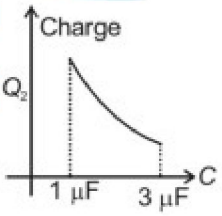
18. In the given circuit, charge Q_2 on the $2\mu F$ capacitor changes as C is varied from $1\mu F$ to 3 μFQ_2 as a function of C is given properly by (Figures are drawn schematically and are not to scale)



A) 

*B) 

C) 

D) 

19. A charge 1C mass 1kg is thrown at $\sqrt{3}\,m/c$ a uniform magnetic field of intensity $\frac{2}{\pi}\times 10^7\,A/m$ at angle 60^0 with it. The radius of curvature of the path followed by it will be

A) 0.25 m B) $\frac{\sqrt{3}}{8}m$ C) 0.5 m D) Zero

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20. For an RLC circuit driven with voltage of amplitude v_m and frequency $\omega_0 = \frac{1}{\sqrt{LC}}$ the current exibits resonance. The quality factor, Q is given by

A) $\frac{\omega_0 L}{R}$ B) $\frac{\omega_0 R}{L}$ C) $\frac{R}{(\omega_0 C)}$ D) $\frac{CR}{\omega_0}$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

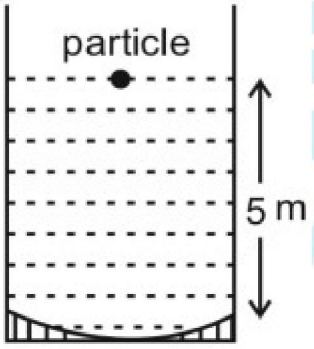
This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. A paramagnetic sample shows a net magnetisation of 6 A / m when it is placed in an external magnetic field of 0.4 T at a temperature of 4 K. When the sample is placed in an external magnetic field of 0.3 T at a temperature of 24 K, then the magnetisation will be ____ $(\text{in } mA/m)$

22. A bar magnet is demagnetized by inserting it inside a solenoid of length 0.2 m,100 turns, and carrying a current of 5.2 A . The coercivity of the bar magnet is __(in SI units)

23. A series AC circuit containing an inductor (20 mH), a capacitor $(120\mu \text{ F})$ and a resistor (60Ω) is driven by an AC source of 24 V / 50 Hz . The energy dissipated in the circuit in 60s is ____ (J)

24. A concave mirror has radius of curvature of 40 m . It is at the bottom of a glass that has water filled up to 5m (see figure). If a small particle is floating on the surface of water, its image as seen, from directly above the glass, is at a distance d from the surface of water. The value of d is close to (Refractive index of water $=1.33$)_____ (in cm)



25. If, for an n-type semiconductor, the density of electrons is 10^{19} m^{-3} and their mobility is 1.6 $m^2/(V.s)$ then the resistivity of the semiconductor is close to _____ $(\text{in } 10^{-1}\,\Omega - m)$

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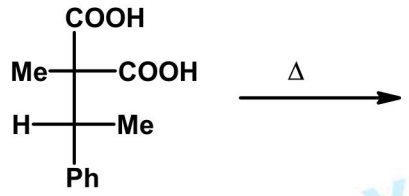
CHEMISTRY **MAX.MARKS: 100**

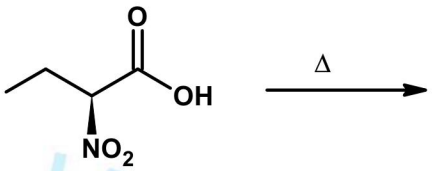
SECTION – I
(SINGLE CORRECT ANSWER TYPE)

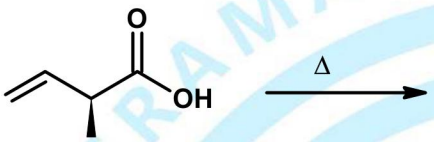
This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

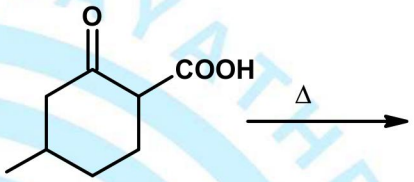
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

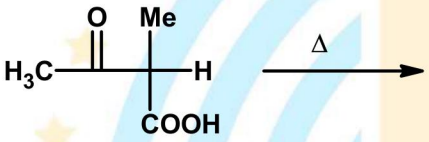
26. Which of the following reaction(s) produce diastereomers as product mixture:

P) 

Q) 

R) 

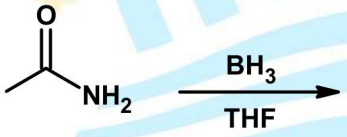
S) 

T) 

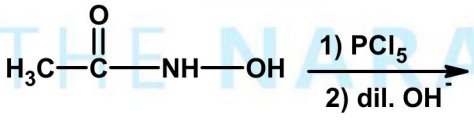
A) P, Q, S only B) only P

C) Q, R, S, T only D) P, Q, T only

27. In which of the following reactions, primary amine will not be formed:

A) 

B) 

C) 

D) 

28. An organic compound 'P' with molecular formula $C_9H_{10}O$ gives the following chemical tests:

(a) undergoes canizzaro reaction

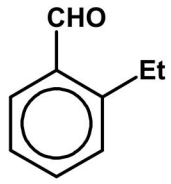
(b) on vigorous oxidation, pthalic acid is obtained

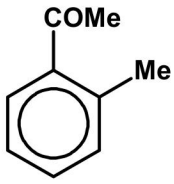
(c) forms a 2,4-DNP derivative

(d) reduces tollen's reagent.

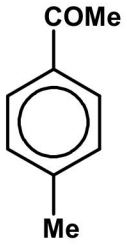
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Then 'p' is:

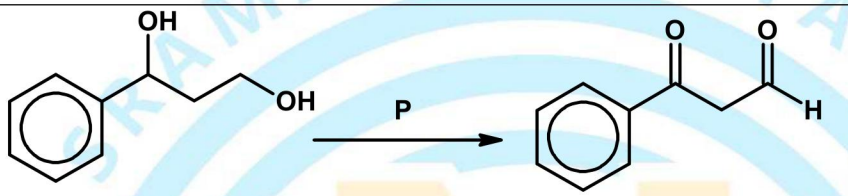
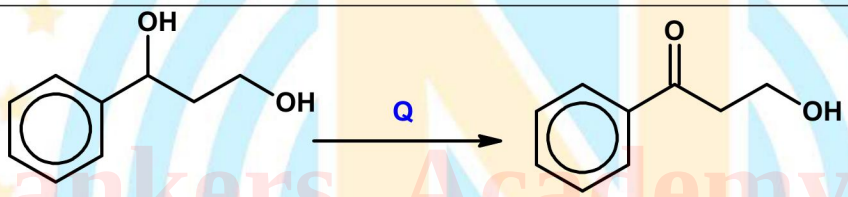
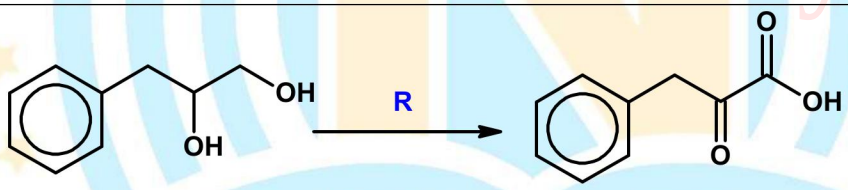
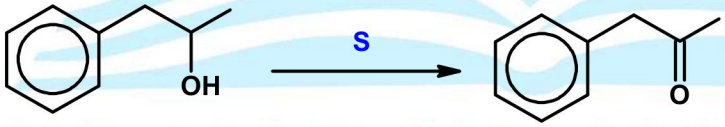
A) 

B) 

C) 

D) 

29. Study the following reactions and match the oxidising agents in list-II with the reactions in List-I:

	Column-I		Column-II
a)		1)	Collins reagent
b)		2)	PCC
c)		3)	MnO_2
d)		4)	Chromic acid
		5)	CrO_3 / $H^{\oplus}$

A) P-1,3; Q-4,5; R-2; S-4

B) P-1,2; Q-3; R-4,5; S-1,2,4,5

C) P-1,3; Q-4; R-4,5; S-3

D) P-2; Q-4; R-1,3; S-4,5

For More Join: @IITJEE_Advanced

30. Match the following.

	Column-I		Column-II
P)	Glycosidic linkage	1)	Globular protein
Q)	Maltose	2)	Connects two monosaccharide units
R)	Peptide bond	3)	Monomeric unit of nucleic acids
S)	Nucleotide	4)	Connects two amino acid units

A) P-4; Q-3; R-2; S-1 B) P-3; Q-4; R-1; S-2

C) P-2; Q-1; R-4; S-3 D) P-1; Q-2; R-3; S-4

31. The chirality in DNA & RNA molecules is due to:

A) D-sugar component B) L-sugar component

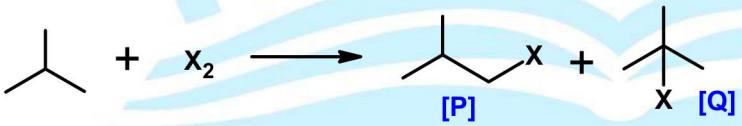
C) chiral bases D) chiral phosphate ester groups

32. Which among the following pairs of vitamins are stored in our body for longer duration?

A) Vitamin 'A' & Vitamin 'D' B) ascorbic acid & Vitamin 'D'

C) Thiamine and ascorbic acid D) Thiamine and vitamin 'A'

33. Choose the incorrect statement regarding the following reaction:



A) [P] is major product when 'X ' is chlorine.

B) [Q] is formed almost exclusively as the product when 'X' is Bromine.

C) when 'X' is chlorine, activation energy for abstraction of primary hydrogen is more than that for abstraction of tertiary hydrogen

D) Reaction is controlled by probability factor when 'X' is chlorine.

For More Join: @IITJEE_Advanced

34. Acetaldehyde will test positive in how many of the following tests.

i) Lucas test ii) Victor mayor test

iii) Iodoform test iv)Sodium bisulfite test

v) litmus test vi) sodium metal test

vii) Tollen's test viii) Fehling's solution test

ix) 2,4-DNP test

A) 4 B) 6 C) 5 D) 3

35. The number of neutrons present in the more abundant isotope of boron is 'x'. Amorphous boron upon heating with air forms a product, in which the oxidation state of boron is 'y'. The value of x + y is …

A) 4 B) 6 C) 3 D) 9

36. Which of the following IUPAC name with respect to given compound is False

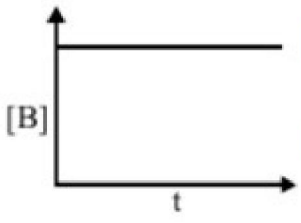
A) $[Pt(Py)_4][PtCl_4]$ – Tetrapyridineplatinum(II) tetrachloridoplatinate(II)

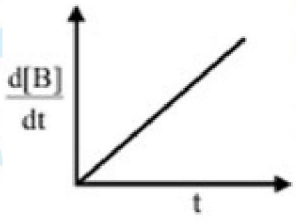
B) $[Cr(en)_3]Cl_3$ – Tris(ethylenediamine)chromium(III) chloride

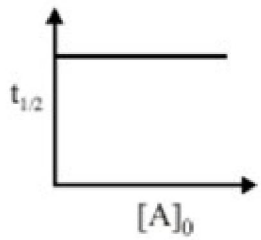
C) $\left[CuCl_2\left(CH_3NH_2\right)_2\right]$ – Dichloridobis(methylamine)copper(II)

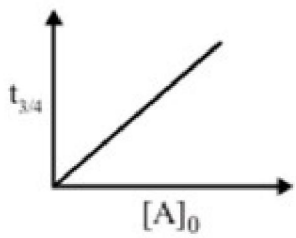
D) $\left[Pt\left(NH_3\right)_2Cl\left(NO_2\right)\right]$ – Diamminechloridonitrito-N-Platinate(II)

37. Which graph represents zero order reaction $\left[A(g) \rightarrow B(g)\right]$:

A) 

B) 

C) 

D) 

For More Join: @IITJEE_Advanced

38. Which halogen is known to cause the reaction given below:

$2Cu^{2+} + 4X^{-} \rightarrow Cu_2X_2(s) + X_2$

A) Only Iodine B) Only Bromine

C) All halogens D) Only Chlorine

39. The nature of oxides V_2O_3 and CrO is indexed as 'X' and 'Y' type respectively. The correct set of X and Y is:

A) X = acidic Y = acidic B) X = basic Y = amphoteric

C) X = basic Y = basic D) X = amphoteric Y = basic

40. Which one of the following complexes will exhibit the least paramagnetic behaviour?

[Atomic number, Cr = 24, Mn = 25, Fe = 26, Co = 27]

A) $\left[Cr(H_2O)_6\right]^{2+}$ B) $\left[Fe(H_2O)_6\right]^{2+}$

C) $\left[Co(H_2O)_6\right]^{2+}$ D) $\left[Mn(H_2O)_6\right]^{2+}$

41. The Crystal Field Stabilization Energy (CFSE) magnetic moment (spin-only) of an octahedral aqua complex of a metal ion $\left(M^{z+}\right)$ are $-0.8\ \Delta_0$ and 3.87 BM, respectively. Identify $\left(M^{Z+}\right)$:

A) V^{3+} B) Co^{2+} C) Cr^{3+} D) Mn^{4+}

42. Match the property given in Column I with the element given in Column II.

	Column-I (Property)		**Column-II** (Element)
A)	Lanthanoid which shows +4 oxidation state	I)	Pm
B)	Lanthanoid which can show +2 oxidation state	II)	Ce
C)	Radioactive lanthanoid	III)	Lu
D)	Lanthanoid which has $4f^7$ electronic configuration in +3 oxidation state	IV)	Eu
E)	Lanthanoid which has $4f^{14}$ electronic configuration in +3 oxidation state	V)	Gd

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A) (A) - II; (B) - IV; (C) - I; (D) - V; (E) - III

B) (A) - I; (B) - II, III; (C) - IV; (D) - I, V; (E) -V

C) (A) - IV; (B) - V; (C) - I, II; (D) - I; (E) - III

D) (A) - I; (B) - IV; (C) - I, II; (D) - I; (E) - III

43. In which of the following options all species contain $X-O-X$ bonds in structure

($X=$ central atom)

A) $H_2S_2O_5$, S_3O_9, $S_2O_6^{-2}$

B) $P_4O_{10}, P_4O_6, H_3P_3O_9$

C) N_2O_5, N_2O, N_2O_4

D) $H_4P_2O_7, H_4P_2O_6, H_4P_2O_5$

44. Which of the following order is not correct :-

A) $CO_2 < SiO_2 < GeO_2 < SnO_2 < PbO_2$ (Oxidising nature)

B) $MnO_4^- > TeO_4^- > ReO_4^-$ (Oxidising nature)

C) $CH_4 < SiH_4 < GeH_4 < SnH_4 < PbH_4$ (Reducing nature)

D) $HOCl < HClO_2 < HClO_3 < HClO_4$ (Oxidising nature)

45. The green colour produced in the borax bead test of a chromium (III) salt is due to the formation of?

A) $Cr(BO_2)_3$ B) CrB C) $Cr_2(B_4O_7)_3$ D) Cr_2O_3

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. The vapour pressure of pure benzene and methyl benzene at 27^0 C is given as 80 Torr and 24 Torr, respectively. The mole fraction of methyl benzene in vapour phase, in equilibrium with an equimolar mixture of those two liquids (ideal solution) at the same temperature is ___ $\times 10^{-2}$ (nearest integer)

For More Join: @IITJEE_Advanced

47. The freezing point depression of a 0.1 molal aqueous solution of an acid H_xA is $0.26\,^0C$. If the acid is 20% dissociated find the value of x . Given $K_f = 1.86 kkgmol^{-1}$ and molality is equal to molarity.

48. A flask contains a mixture of compounds A and B. Both compounds decompose by first-order kinetics. The half-lives for A and B are 300 s and 180 s, respectively. If the concentrations of A and B are equal initially, the time required for the concentration of A to be four times that of B(in s) is_____ (Use ln 2 = 0.693)

49. Number of oxygen atoms present in chemical formula of fuming sulphuric acid is_____

50. The number of primary amides (benzenoid) with molecular formula C_8H_9NO which on reaction with Br_2 / NaOH gives primary amine as major product are _____

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MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. If a set of events $E_1, E_2, \cdots . E_n$ represent a partition of the sample space S, then which of the following is/are not a necessary condition

$C-1\ \mathrm{E}_i \cap \mathrm{E}_j = \phi, i \neq j, i, j = 1,2,3,\ldots,n$

C-2 $\mathrm{E}_1 \cup \mathrm{E}_2 \cup \ldots \cup \mathrm{E}_n = \mathrm{S}$

C-3 $\mathrm{P}(\mathrm{E}_i) > 0$ for all $i = 1,2,\ldots,n$.

C-4 $E_1, E_2, E_3, \cdots E_n$ are equi probable

A) $C-1$ is not necessary B) $C-3$ and $C-2$ not necessary

C) Only $C-3$ is not necessary D) Only $C-4$ is not necessary

52. Let slope of a tangent at point P on the ellipse $\frac{x^2}{81}+\frac{y^2}{9}=1$ is m such that normal at P is at a maximum distance from origin, then $|3\ \mathrm{m}|$ is equal to

A) 1 B) $\sqrt{3}$ C) $\sqrt{2}$ D) $\frac{1}{\sqrt{3}}$

53. The minimum value of

$$f(x)=\frac{(x-2)^{10}+(x-2)^{11}+3(x-2)^{12}+(x-2)^{15}+(x-2)^{23}+(x-2)^{25}}{(x-2)^{15}} \forall x > 2$$

A) 2 B) 4 C) 6 D) 8

54. Let $\mathrm{S} \equiv 0$ and $\mathrm{S}' \equiv 0$ be two circles belonging to the family of circles $\mathrm{x}^2+\mathrm{y}^2+\lambda \mathrm{x}-4\mathrm{y}+3=0 : (\lambda \in \mathrm{R})$ such that $\mathrm{S} \equiv 0$ bisects the circumference of circle $\mathrm{S}' \equiv 0$ and has radius equal to twice of radius of $S' \equiv 0$. Then distance between their centres is

A) $\frac{1}{2}$ B) 1 C) 2 D) $\frac{1}{\sqrt{2}}$

For More Join: @IITJEE_Advanced

55. Consider the parabola $E_1 : y^2 = 8x$ and ellipse $E_2 : x^2 + 4y^2 - 4k = 0$. If a common tangent to E_1 and E_2 meets them at points A and B respectively. If eccentric angle of point B is $\frac{2\pi}{3}$; then the value of k is equal to _____

A) 18 B) 24 C) 30 D) 36

56. Let $P(a,b)$ be a point on the curve $y = 2x^2 + 6x + 5$ nearest to the line $2x + y + 3 = 0$, then the equation of the normal to the curve at P is

A) $x + 2y - 4 = 0$ B) $2x + y + 4 = 0$

C) $2x - y + 3 = 0$ D) $x - 2y + 4 = 0$

57. The number of ways of arranging the letters of the word ALGEBRA in which consonants appear in alphabetical order is _____

A) 35 B) 105 C) 70 D) 90

58. Abhinav flips 5 fair coins exactly three of which land heads, the probability the first two are both heads is_____

A) $\frac{4}{5}$ B) $\frac{3}{10}$ C) $\frac{1}{5}$ D) $\frac{3}{5}$

59. Let $f(x) = -x^2 + 10x - 20$, value of $\int_0^5 f(f(f(x)))dx$ is _____.

A) $5\left(1 - \frac{5^7}{9}\right)$ B) $5^3\left(1 - \frac{5^7}{9}\right)$ C) $5^2\left(1 - \frac{5^7}{9}\right)$ D) $5^4\left(1 - \frac{5^7}{9}\right)$

60. Which of the following statements is/are True?

S1: If A and B are two events such that $P(A) \neq 0$ and $P(B/A) = 1$ then $A = B$

S2: If $P(A/B) > P(A)$ then $P(B/A) > P(B)$

S3: If A and B are two events such that $P(A) + P(B) - P(A \cap B) = P(A)$ then $P(A/B) = 1$

A) Only S2 is True B) S2 and S3 True, S1 is false

C) only S1 is True D) All are True

For More Join: @IITJEE_Advanced

61. Sum of positive solutions to $2x^2 - x[x] = 5$, where [x] is largest integer less than or equal to x is _____

A) $\frac{3+2\sqrt{41}-2\sqrt{11}}{4}$ B) $\frac{\sqrt{41}+\sqrt{11}}{2}$

C) $\frac{3+2\sqrt{41}+2\sqrt{11}}{4}$ D) $\frac{3+\sqrt{41}+2\sqrt{11}}{4}$

62. Evaluate

$$\int \sec^3 2x \cdot \sin^4 3x \cdot e^{5x}\left(6\tan 2x + 12\cot 3x + 5\right)dx$$

A) $\sec^3 2x \cdot \sin^4 3x \cdot e^{5x} + C$ B) $\sec^3 2x \cdot \sin^3 3x \cdot e^{5x} + c$

C) $\sec^2 2x \cdot \sin^3 3x \cdot e^{5x} + c$ D) $\sec^2 2x \cdot \sin^3 3x \cdot e^{5x} + c$

63. How many statements of following is/are **TRUE**?

S1 : Consider four distinct points $M_i(z_i)$, $i \in \{1,2,3,4\}$, the lines M_1M_2 and M_3M_4 are orthogonal if and only if $\frac{z_1 - z_2}{z_3 - z_4} \in i\mathbb{R}^*, \mathbb{R}^* = \mathbb{R} - \{0\}, i = \sqrt{-1}$

S2 : Distinct points $M_1(z_1), M_2(z_2), M_3(z_3), M_4(z_4)$ are concyclic if and only if

$$\frac{z_3 - z_2}{z_1 - z_2} : \frac{z_3 - z_4}{z_1 - z_4} \in R$$

S3 : if distinct points $M_1(z_1), M_2(z_2), M_3(z_3), M_4(z_4)$ are concyclic then

$$\frac{z_3 - z_2}{z_1 - z_2} : \frac{z_3 - z_4}{z_1 - z_4} \in R$$

S4: if distinct points $M_1(z_1), M_2(z_2), M_3(z_3), M_4(z_4)$ are collinear then

$$\frac{z_3 - z_2}{z_1 - z_2} : \frac{z_3 - z_4}{z_1 - z_4} \in R$$

A) One B) Two C) Three D) Four

For More Join: @IITJEE_Advanced

64. Two important properties of Definite integrals are listed below:

(1) Second Fundamental Theorem of Integral Calculus

Let f be continuous function defined on the closed internal $[a,b]$ and F be an antiderivative of f, then $\int_a^b f(x)dx = F(b)-F(a)$

(2) $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx : c \in (a,b)$

By using these properties.

Evaluate $\int_{-1}^{3/2}\left(\left|x\sin(\pi x)\right|dx + \left[x^3\right]\right)dx$ $[\cdot]$ *is G.I.F* is

A) $\frac{3}{\pi}+\frac{1}{\pi}-\sqrt[3]{2}-\sqrt[3]{3}$

B) $\frac{3}{\pi}+\frac{1}{\pi^2}+\frac{5}{2}-\sqrt[3]{2}-\sqrt[3]{3}$

C) $\frac{3}{\pi}-\frac{1}{\pi^2}+\frac{5}{2}-\sqrt[3]{2}-\sqrt[3]{3}$

D) $\frac{3}{\pi}+\frac{1}{\pi^2}+\frac{5}{2}-\sqrt[3]{2}+\sqrt[3]{3}$

65. Let $I=\int_0^{\sqrt{2}} e^{x^2}dx$, $J=\int_0^{\pi/2}\sqrt{\sin x}dx$ then

A) $I<\frac{9}{2}, J<\frac{\pi}{4}$

B) $I<\frac{9}{2}, J>\frac{\pi}{4}$

C) $I>\frac{9}{2}, J>\frac{\pi}{4}$

D) $I>\frac{9}{2}, J<\frac{\pi}{4}$

66. Differential equation of the family of circles in the second quadrant and touching the co-ordinate axes is

A) $(x+y)^2\left[\left(\frac{dy}{dx}\right)^2+1\right]=\left[x+yy'\right]^2$

B) $(x+y)^2\left[\frac{dy}{dx}+1\right]=\left[x+yy'\right]^2$

C) $(x+y)\left[\left(\frac{dy}{dx}\right)^2+1\right]=\left[x+yy'\right]^2$

D) $(x+y)^2\left[\left(\frac{dy}{dx}\right)^2+1\right]=\left[x+yy'\right]$

For More Join: @IITJEE_Advanced

67. If $\frac{dy}{dx}+2y\tan x=\sin x, 0<x<\frac{\pi}{2}$ and $y\left(\frac{\pi}{3}\right)=0$, then area bounded by $y=y(x), x=0$, $y=0 \& x=\frac{\pi}{3}$ is ____

A) $-\frac{\pi}{3}+\frac{\sqrt{3}}{4}$ B) $\frac{\pi}{3}+\frac{\sqrt{3}}{4}$ C) $\frac{\pi}{3}-\frac{\sqrt{3}}{4}$ D) $\frac{\sqrt{3}}{4}+\frac{\pi}{2}$

68. If $\frac{3+i\sin\theta}{4-i\cos\theta}$, $\theta\in[0,2\pi]$, is a real number, then an argument of $\sin\theta+i\cos\theta$ is

A) $\pi-\tan^{-1}\left(\frac{4}{3}\right)$ B) $\pi-\tan^{-1}\left(\frac{3}{4}\right)$ C) $-\tan^{-1}\left(\frac{3}{4}\right)$ D) $\tan^{-1}\left(\frac{4}{3}\right)$

69. All the points in the set $S=\left\{\frac{\alpha+i}{\alpha-i}, \alpha\in R\right\}, i=\sqrt{-1}$ lie on a

A) straight line whose slope is -1

B) circle whose radius is $\sqrt{2}$

C) circle whose radius is 1

D) straight line whose slope is 1

70. The coefficient of x^{2012} in the expansion of $(1-x)^{2008}(1+x+x^2)^{2007}$ is equal to

A) 3 B) 2 C) 1 D) 0

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
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71. If $\alpha+\frac{1}{\alpha}$ and $2-\beta-\frac{1}{\beta}(\alpha,\beta>0)$ are the roots of the quadratic equation $x^2-2(a+1)x+a-3=0$, then the sum of all integral values of a is ____

72. The chord AB of the circle $x^2+y^2=8$ is given by $mx-y=2m-2; m\in R$. If P is a variable point on the circle such that $\angle APB=\frac{\pi}{4}$; then length of chord AB is _____

For More Join: @IITJEE_Advanced

73. Let $H: xy - y - 2x - 2 = 0$. If normal drawn to H at $P(3,4)$ intersects the curve again at $Q(\alpha+1, \beta+1)$, then $\frac{\alpha}{\beta}$ is

74. Number of 5-digit natural numbers such that the product of their digits equals number of ways to create a necklace out of 6 beads from 10 distinct beads is

[Clockwise & Anticlockwise are same]

75. Let a three term strictly increasing arithmetic sequence of integers are one such that the sum of the squares of the three terms equals the product of the middle term and the square of the common difference, then the sum of all possible values of third term of such sequence is 'p', value of 'p' is____

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